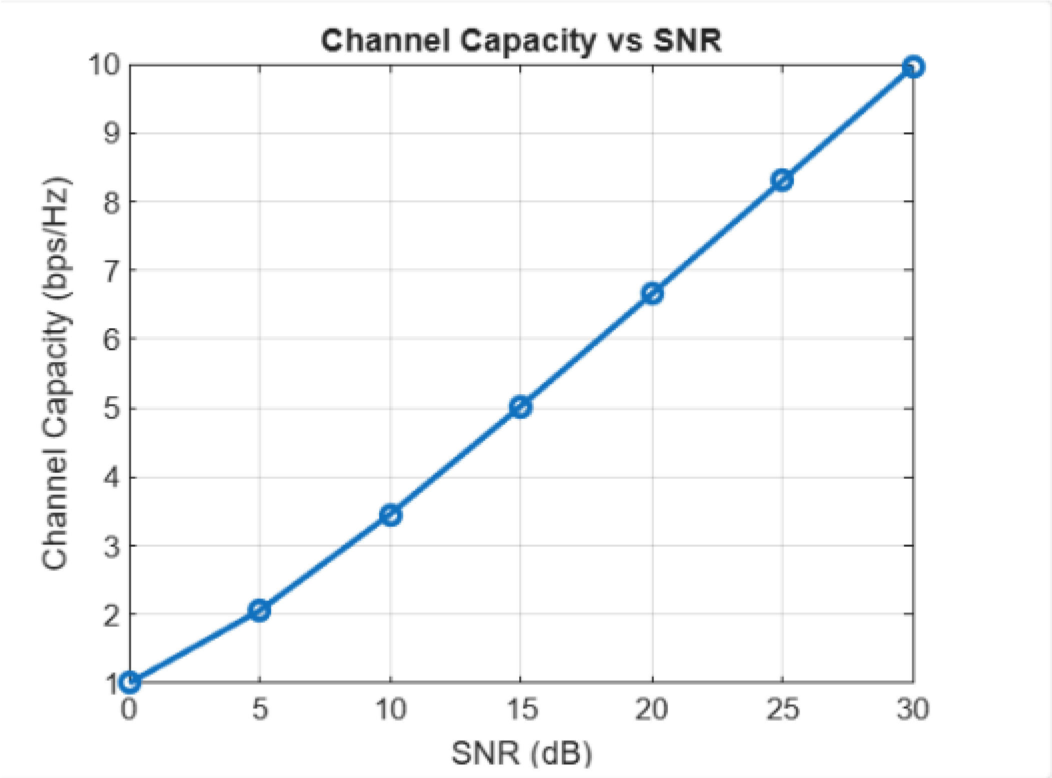
Experiment: 9

Massive MIMO Capacity vs SNR and Gain Using MATLAB

Objective – 1 Plot of Channel Capacity (bps/Hz) versus SNR (dB).

```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```



Objective – 2 Bar graph comparing SNR values under different communication scenarios.

```
clc;
clear;
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snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```

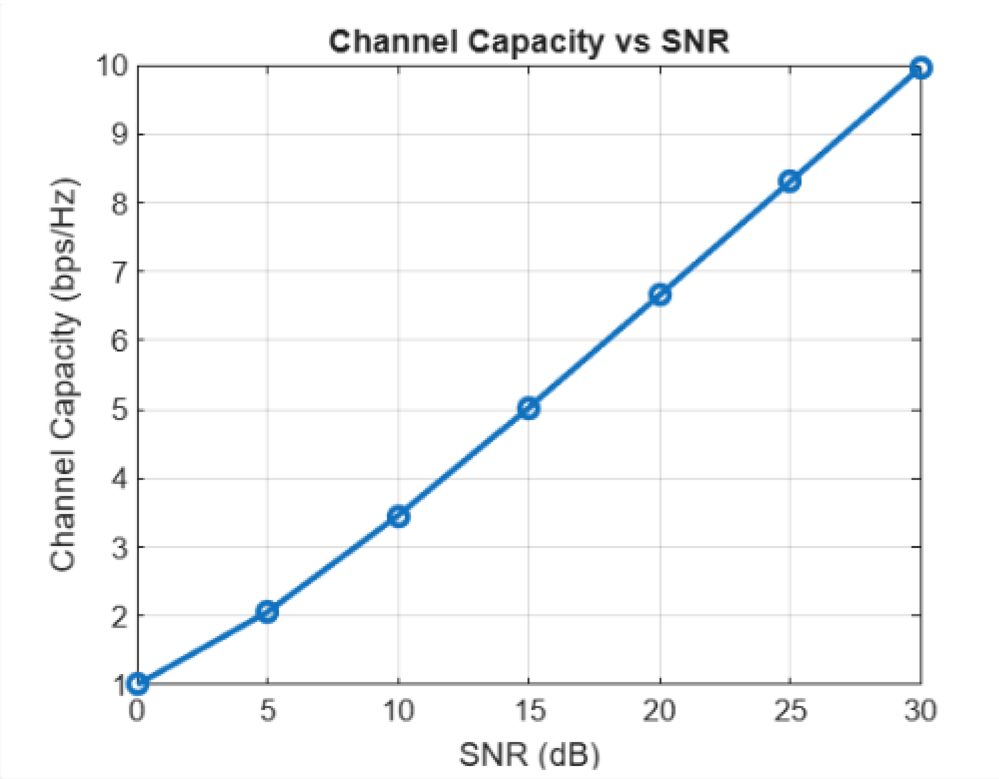

Objective – 3 Bar graph showing Antenna Gain for different antenna array sizes.

```
clc;
clear;
close all;

% Number of Antenna Elements
antennas = [4 8 16 32 64];

% Antenna Gain (dB)
gain = 10 * log10(antennas);

% Create Bar Graph
figure;
bar(antennas, gain);

grid on;
xlabel('Number of Antenna Elements');
ylabel('Antenna Gain (dB)');
title('Antenna Gain for Different Antenna Array Sizes');
```